

The session on Computer Algebra for Dynamical Systems and Celestial Mechanics

Santiago de Compostela, Spain , June 18-22, 2018

Organizers

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Overview

Celestial Mechanics and Dynamical Systems are traditional fields for applications of computer algebra. Computer algebra methods play a fundamental role in the treatment of concrete problems and applications. Computer algebra applications include nontrivial use of existing systems Maple, Mathematica, Singular etc. and the development and implementation of new algorithms, and specialized packages. The session will bring together specialists from diverse areas: differential equations, dynamical systems and computer algebra. In particular, the following topics will be considered

- Stability and bifurcation analysis of dynamical systems
- Construction and analysis of the structure of integral manifolds
- Symplectic methods
- Symbolic dynamics
- Celestial mechanics and stellar dynamics, N-body problem, KAM theory
- Specialized computer algebra software for celestial mechanics
- Normal form theory and formal integrals
- Deterministic chaos in dynamical systems
- Families of periodic solutions
- Perturbation theories and reductions
- Exact solutions and partial integrals
- Analysis and blow-ups of non-elementary stationary points
- Analysis of singularities: geometry and topology
- Integrability and nonintegrability, algebraic invariant sets and Darboux integrability
- Discrete dynamical systems and ergodic theory
- Topological structure of phase portraits and computer visualization